Multidomain Constrained Translation Network for Change Detection in Heterogeneous Remote Sensing Images

Haoran Wu, Jie Geng, *Member, IEEE*, and Wen Jiang

***Abstract*— In heterogeneous image change detection (HICD), preventing neural networks from distorting critical information is the main challenge of such methods based on deep translation. Most of these methods rely on a priori information to suppress the effects of changed pixels in the translation process, but the accuracy of the prior information will influence the results of translation. In this article, we propose an end-to-end multidomain constrained translation network (MDCTNet) for unsupervised HICD. The proposed MDCTNet utilizes an improved generative adversarial network (GAN) to generate target domain images realistically. Furthermore, to retain the content information of the source domain images, MDCTNet leverages contrastive learning to ensure the consistency of adjacent pixel relationships. Meanwhile, it employs high-frequency information consistency which preserves pivotal characteristics. We compare the proposed MDCTNet with state-of-the-art algorithms to verify the efficacy of the proposed technique. The experimental results on five real datasets demonstrate the effectiveness of the proposed method.**

***Index Terms*— Contrastive learning, global-local constraint, heterogeneous image change detection (HICD), remote sensing, spatial-frequency domain constraint.**

I. Introduction

CHANGE detection is a fundamental application in the realm of remote sensing, with the specific task of identifying changes occurring on Earth by interpreting two or more images of the same geographical location captured at varying instances in time [1]. It has been a hot topic in remote sensing image processing, and with the development of remote sensing technology, the ever-increasing deluge of remote sensing data necessitates urgent processing. Change detection techniques are widely used in problems such as urban development studies, land use monitoring, disaster assessment (earthquakes, fires, and floods) and environmental research (e.g., deforestation, lake glacier reduction). Due to the rapid progress of artificial intelligence technology, deep learning methods for the interpretation of remote sensing imagery are gaining substantial attention in the research community. Over the last few years, there have been multiple reviews [2], [3], [4], [5] discussing the topic of change detection in remote sensing images using deep learning algorithms.

Recent years have witnessed a growth in the number of proposed change detection techniques for remote sensing images, with novel approaches continuing to emerge. It is worth mentioning that with the development of artificial intelligence technology, change detection methods based on deep learning have received wide attention and achieved good performance in remote sensing [6], [7], [8], [9]. However, most of the previous change detection methods rely on homogeneous data, i.e., multitemporal remote sensing data acquired by the identical sensor or the sensor belonging to a similar category and sharing the same acquisition parameters [10], [11], [12], [13], [14], [15]. Such an ideal scenario guarantees that the change detection technique is exclusively concentrated on the portion of the monitored area that undergoes genuine variations, while disregarding the impacts of disparities in the collection of data. As remote sensing imaging technology continues to advance, increasing numbers of multisource remote sensing data are now available, which cover the same geographic region, such as synthetic aperture radar (SAR) image, geographic information system (GIS) data, high-resolution satellite images, and various spatio-temporal measurements [2], [16], [17]. In addition, relying exclusively on a specific data source restricts the applications of change detection algorithms, particularly concerning their ability to detect and respond promptly to unforeseen events and to provide long-term monitoring of ground conditions. Therefore, it becomes more and more urgent to develop heterogeneous image change detection (HICD) techniques to meet the challenges of future multisource data.

The HICD methods utilize images from multiple sensors, thus they have well-application to real-world scenes. Nevertheless, these approaches present further obstacles since the utilization of images acquired from distinct sensors may entail varying spatial resolutions, spectral channels, wavelength ranges, and radiometric resolutions. And the analysis of diverse remote sensing images obtained through varying seasonal or lighting conditions, using a homogeneous approach, can lead to substantial inaccuracies. The former represents strong heterogeneity, while the latter represents mild heterogeneity, and we believe that as long as the data are not completely homogeneous, then they should be considered as heterogeneous data. The summary is that the main difficulty

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in HICD lies in the fact that heterogeneity implies different data distributions and an unbalanced amount of information between two images [18]. This is especially evident when it comes to heterogeneous sensors because they have different imaging mechanisms and large differences in imaging results [19]. Therefore, if the data is not preprocessed, using homogeneous algorithms directly will bring significant errors. To address this challenge, many researchers have explored many methods to achieve HICD. First, some traditional methods are presented, [20], [21], [22], [23], these methods concentrate on assessing similarities between individual pixel points of heterogeneous image pairs and utilize the correlation or mutual information of image patches as a gauge to quantify the degree of change in the images. Wan et al. [24] proposed a post-classification comparison (PCC) strategy that derives change outcomes through the comparison of classification maps of heterogeneous remote sensing images. Nevertheless, the method's efficacy relies considerably on the precision of the image classification algorithm employed. Liu et al. [25] used the uniform pixel transform to transform the heterogeneous images into a common space to facilitate change detection. Luppino et al. [26] employed image regression to map images from a source domain to a target image domain, with pseudo-training data generated from a priori affinity matrices. Mignotte [27] used a modified geometric fractal decomposition and shrinkage mapping method to project images to any image modality.

There are also a range of examples of HICD methods based on deep learning. Saha et al. [28] employed deep clustering and contrastive learning to train the network in a self-supervised manner to obtain change information for multimodal diachronic image pairs. Zhang et al. [29] proposed a style-based recalibration module to efficiently capture seasonal features in order to address the impact of seasonal changes. The three methods described in [30], [31], and [32] adopt the principle of deep translation, wherein remote sensing images displaying diverse features are converted into a unified feature space. The resultant feature maps are then utilized to extract information for discriminating changes by comparing them directly. While equally inspired by the concept of image translation, the techniques presented in [33], [34], and [35] set themselves apart from the preceding three methods as they transform the source domain image to the target domain image, consequently creating a dual-time image pair in each domain. This type of approach obtains the change information and aggregates it in both domains separately, making the final result more reliable.

In general, the practice of collecting images from disparate domains and aligning them for comparison using deep translation techniques is widely adopted, and our proposed approach applies this same principle. In carrying out HICD, our objective is to ensure that the output can replicate the appearance of the target domain image. Notably, to avoid any loss of important information, preserving specific source domain image's content or structure is critical. A number of depth translation-based techniques currently maintain content through cycle consistency, which, while favorable, hamstrings the technique by creating assumptions of bijectivity between two domains [36]. Consequently, it is imperative to investigate alternative strategies for implementation. Additionally, some techniques rely on a priori information to suppress the effects of changed pixels in the translation process, but the accuracy of the prior information may influence the results of translation. We introduce a multidomain constrained translation network (MDCTNet) for HICD to preserve the source domain images' structure while generating pseudo-target domain images whose style is as consistent as possible with the reference. The proposed MDCTNet employs a completely end-to-end manner of source domain to target domain translation without any need for supervision or prior knowledge. Given the intricate ground coverage exhibited by remote sensing images, the problem of retaining the content of multiscale objects needs to be addressed. Existing translation-based methods employed for natural images become impractical for remote sensing images when applied directly. In this research, a novel approach is introduced for remote sensing image translation. The proposed method leverages contrastive learning and high-frequency information constraint for effectively constraining the translation at both global and local scales. The main contributions of this article are summarized as follows.

1) We propose an unsupervised MDCTNet, aimed at addressing the challenges posed by heterogeneity through an end-to-end framework, which is effective in reducing the number of false positives (FPs) in HICD results.
2) We design a multiscale discriminator to factor in the complex coverage of remote sensing images, which can force the model to generate stylized images from coarse to fine.
3) We propose contrastive constraint loss and frequency domain constraint loss. These losses work in tandem to strike an effective balance between image translation and the retention of critical structures within the image.

This article is structured as follows: Section II provides an overview of the related work. Section III outlines the proposed MDCTNet for HICD. Section IV presents an analysis of the performance of the proposed method through experiments. Finally, the conclusions are summarized in Section V.

II. Related Work

A. Image-to-Image Translation

Currently, the area of image-to-image translation (I2I) is a prominent field of investigation in the sphere of computer vision. The aim of I2I is to learn a mapping between an input image and a corresponding output image that belongs to a different domain. Recent works in I2I have been primarily focused on developing models that can generate high-quality images with better visual fidelity and perceptual consistency. One approach that has been particularly successful is conditional generative adversarial networks (cGANs) [37], which have shown impressive results in various I2I tasks. However, this method requires paired training data. Several variants of GANs have been proposed to address the limitations of the original framework. For example, CycleGAN [38], which is a popular variant of GANs, is designed for unsupervised I2I

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without relying on a paired training dataset. It is based on cycle consistency loss, where the output image of the translator is expected to be mapped back to the input domain by reverse mapping so that it can reconstruct the original input image.

In addition, various representative studies have been proposed that leverage cycle consistency loss as an effective means to generate high-quality images. Kim et al. [39] proposed a new attention mechanism based on CycleGAN with an auxiliary classifier. This mechanism is able to direct the generator to focus on those more important regions that distinguish the source domain from the target domain. StarGAN and its variants [40], [41] achieve migration of multiple domains using only one model, which improves the scalability and robustness of image domain migration. UNIT [42] proposes the assumption that different data spaces share the same hidden space. More recently, in MUNIT [43] the shared hidden space is referred to as content, and the approach also assumes that there should also be a style space that characterizes differences between different data. Stylegan and its variants [44], [45], [46] proposed a style-based generator that implements unsupervised attribute separation. Chen et al. [47] proposed to reuse discriminators to encode images in the target domain and developed a decoupled training strategy. Cai et al. [48] proposed to add constraints in the frequency domain to the existing framework, which can better preserve the content of the input.

In contrast to the aforementioned methods, Park et al. [36] proposed a contrastive learning-based framework and without using cycle consistency. Furthermore, based on the contrastive learning framework, Wang et al. [49] proposed a generator to generate hard negative samples. All of the above work is for natural images, and there is currently some work [50], [51], [52] for translation of remote sensing images.

B. HICD Based on Deep Translation

HICD has garnered significant attention in contemporary times owing to its pivotal contribution to land cover change detection, particularly during instances of natural calamity, where homogeneous images are lacking. In this review [53], it is mentioned that existing HICD methods fall into four main categories: 1) image fusion; 2) local image feature descriptions; 3) methods based on PCC; and 4) image translation. For example, among the image fusion methods, Solarna et al. [54] propose to utilize Markov and conditional random fields to achieve multimodal SAR data fusion techniques for unsupervised change detection. The second category of methods (local image feature descriptions) mentioned above is often defined as traditional algorithms and they are based on the theory that the same ground instances in different images usually have similar structural features [55]. This class of PCC-based methods is applicable for both heterogeneous and homogeneous image change detection because they just need to classify the pixel points in the images of the two moments and compare the classification maps [56]. Since we are focusing on deep translation-based HICD in this article, this class of methods will be described in detail next.

The purpose of the translation is to eliminate the heterogeneity between images and make the diachronic image pairs more comparable. For instance, Jiang et al. [32] proposed a migration learning-based semi-supervised Siamese network (MLSSN) that reduces the learning cost of low-level features while converting heterogeneous pixels to a common domain. Li et al. [34] used a deep translation network (DTCDN) to reduce the difference between optical images and SAR images and then imported the translation results into a supervised change detection network. Recently, the research team of DTCDN has improved DTCDN as MTCDN [57]. MTCDN is a multitask learning framework that combines image translation and change detection tasks. Image translation is responsible for unifying two images into the same feature domain. These two tasks are co-trained and mutually reinforced to improve efficiency and maintain accuracy.

Although the above three methods have achieved good results, they require additional supervised information compared to unsupervised methods, which is usually difficult to obtain. Therefore, the focus of our research is mainly on unsupervised change detection. Liu et al. [30] proposed an unsupervised symmetric convolutional coupled network (SCCN) to convert heterogeneous images into a common feature space and obtain change information by comparing these feature maps. Niu et al. [31] used conditional generation adversarial network (CAN) and an approximation network to convert heterogeneous images to a space with more consistent information representation so that direct comparison can yield change results. Luppino et al. [33] proposed a weighted translation network (X-Net) and an adversarial cyclic encoder network (ACE-Net) to reduce the heterogeneity between images, and then the transformed images are two-by-two for the difference map, and the change detection result map can be obtained after filtered segmentation. Luppino et al. [35] designed a code-aligned autoencoders (CAAs) and the prior information was utilized to suppress the effects of changed pixels in the translation process. UCDFormer [58] employs a lightweight transformer-driven approach for image translation, aiming to alleviate domain shift efficiently in real-time. This unsupervised method primarily addresses variations arising from seasonal and stylistic differences induced by illumination and atmospheric conditions in multitemporal images. For the consideration of the available codes, in this article, we compare our method with these methods, and the detailed comparative analysis is presented in Section IV.

III. Methodology

A. Overview

To solve the problem of not being able to directly compare dual-time image pairs due to image heterogeneity, we propose an MDCTNet for heterogeneous remote sensing image change detection. It mainly consists of two pipelines, deep translation, and change detection, as shown in Fig. 1. In addition, the deep translation network consists of three parts: multiscale discriminator, contrastive constraint, and frequency domain constraint.

Consider a pair of heterogeneous remote sensing images, denoted as $I^{\text{pre}} \subset \mathcal{X}^{H\times W\times C_1}$ and $I^{\text{post}} \subset \mathcal{Y}^{H\times W\times C_2}$, which

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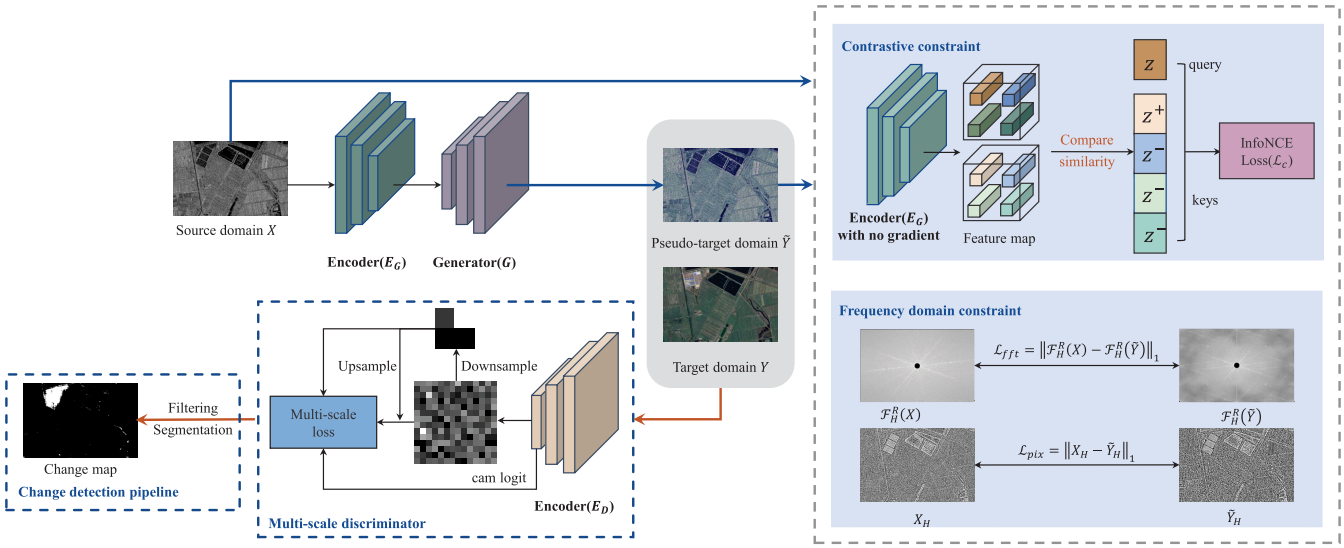


Fig. 1. Flowchart of the proposed MDCTNet. For the given image X and Y, the deep translation network is designed in order to minimize the effect of the heterogeneity between them on the change detection results. The network consists of three main components: a multiscale discriminator, contrastive constraint, and frequency domain constraint. First, we transform the source domain image X into the pseudo-target domain image $\tilde{Y}$ and enhance the quality of the generated image by the multiscale discriminator. Then, we input X and $\tilde{Y}$ into the contrastive constraint module and the frequency domain constraint module, which work in concert to prevent the image from overfitting to the target domain and losing its basic identity features. Finally, we exchange the source and target domains and repeat the process to obtain $\tilde{X}$. By utilizing X, $\tilde{X}$, Y, and $\tilde{Y}$, we are able to perform the next change detection pipeline to obtain the final result.

are, respectively, acquired prior to (prechange) and subsequent to (post-change) a certain event. The ultimate objective is to identify all changes within the scene. Considering the heterogeneity of domains $\mathcal{X}$ and $\mathcal{Y}$, if the data is not preprocessed, using homogeneous algorithms directly will lack meaningful interpretation. Furthermore, assume that $X = I^{\text{pre}} \subset \mathcal{X}^{H\times W\times C_1}$ is the source domain image and $Y = I^{\text{post}} \subset \mathcal{Y}^{H\times W\times C_2}$ is the target domain image. The purpose of the deep translation network is to transform $X \subset \mathcal{X}^{H\times W\times C_1}$ into a pseudo-target domain image $\tilde{Y} \subset \mathcal{Y}^{H\times W\times C_2}$. $\tilde{Y}$ and Y are as similar as possible except for the changed regions. Therefore, this minimizes the effect of the heterogeneity between them on the change detection results.

The translation network consists of three main components: a multiscale discriminator, contrastive constraint, and frequency domain constraint. First, we transform the source domain image X into the pseudo-target domain image $\tilde{Y}$ and enhance the quality of the generated image by the multiscale discriminator. Then, we input X and $\tilde{Y}$ into the contrastive constraint module and the frequency domain constraint module, which work in concert to prevent the image from overfitting to the target domain and losing its basic identity features. Finally, we exchange the source and target domains and repeat the process to obtain $\tilde{X}$. By utilizing X, $\tilde{X}$, Y, and $\tilde{Y}$, we are able to perform the next change detection pipeline. The deep translation network mainly uses GANs to approximate the data distribution making $\tilde{Y} \approx Y$. Compared with existing I2I networks based on cycle-consistency loss, the network proposed in this article only needs to learn mappings in one direction per training with the help of contrastive constraint and it avoids the use of an inverse auxiliary generator and discriminator. Simultaneously training two GAN can be a challenging task, and our proposed approach aims to reduce its complexity by breaking down the cyclic training task without relying on cycle consistency. Our approach offers an intuitive solution to mitigate the difficulties often encountered when training GAN in a joint manner, ultimately improving the training efficiency and effectiveness. After the image translation is completed, the change detection results in domian $\mathcal{Y}$ are obtained by directly comparing the differences between $\tilde{Y}$ and Y.

B. Multiscale Discriminator

As shown in Fig. 1, the complexity of land cover types in remote sensing images makes it a major test to assess the quality of the generated images more effectively. Therefore, we propose a multiscale discriminator that can assess the quality of generated images $\tilde{Y}$ with fine-grained details so as to account for the presence of intricate objects more effectively. The discriminator, denoted by D, comprises two essential components, namely the encoder E_D and the classifier C. The structure of the encoder E_D is shown in Fig. 2, which contains two parts of the output: one is the cam logit, and the other is the attention-weighted feature map. As shown in Fig. 3, to achieve multiscale paradigm, classifier C is partitioned into three subclassifiers, namely D_1, D_2, and D_3, where D_1 is connected from the cam logit in the E_D. Then, a downsampling-convolution layer is applied to the output of the E_D to obtain a smaller-scale feature map, which is connected to two branches: these two branches adopt a U-Net [59] form for each input image and the role of the upsampling block is to restore the feature map to the size it was before the two downsamplings. The adversarial loss consists of three parts, D_1, D_2, and D_3. The specific objective

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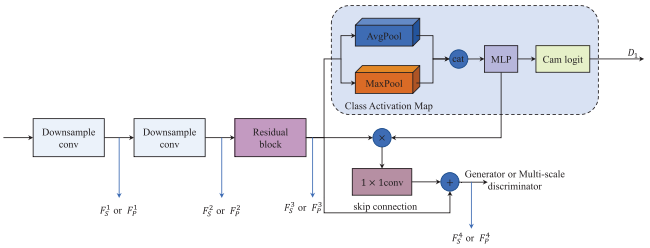


Fig. 2. Structure of the encoder. Encoders E_G and E_D have the same structure. And F_S^i and F_P^i are used as middle layer features of the encoder and they will only be used in the contrastive constraint. In addition, the cam logit output from the above branch will be used as input to D_1 in the multiscale discriminator.

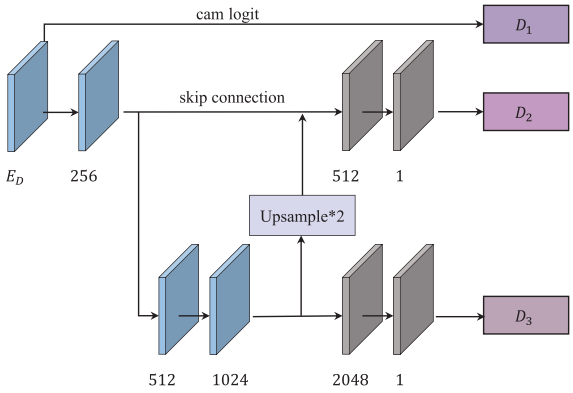


Fig. 3. Structure of the classifier C. C is partitioned into three sub-classifiers, namely D_1, D_2, and D_3, where D_1 is connected from the cam logit in the E_D. The blue blocks in the figure represent downsampling-convolution layers, and the gray blocks represent normal convolutional layers whose role is to reduce the feature map to 1 dimension. In addition, the number below the block represents the number of filters in that convolutional layer.

function takes the following form:

$$\mathcal{L}_{\text{GAN}} = \min_G \max_D \sum_{i=1}^{3} \mathbb{E}_{y\sim Y} \log D_i(y) + \mathbb{E}_{x\sim X} \log(1 - D_i(G(E_G(x)))) \quad (1)$$

where $G(E_G(X)) = \tilde{Y}$, and D_i represents the discriminator at a certain scale.

This approach leverages the advantages of a multiscale network architecture, enabling it to extract multilevel, multicontext features that can measure the fidelity of the generated image more comprehensively. Furthermore, we present two proposed constraints aimed at maintaining the content of the source domain images throughout the image translation process. These constraints serve to address the issue of content preservation, which is a crucial factor in ensuring that the translated images maintain the essential structure of their source domain counterparts. The details of these constraints and their implementation are discussed in Sections III-C and III-D.

C. Contrastive Constraint

Park et al. [36] and Zhu et al. [38] have demonstrated that relying solely on adversarial training may lead to over-adaptation of the images to the target domain, potentially causing the loss of their essential identity characteristics. Therefore, it becomes crucial to address the issue of altering appearance while maintaining the content in image translation. In this article, specifically to keep X and $\tilde{Y}$ with the same content, a simple idea would be to enforce $\min \|X - \tilde{Y}\|_1$, but imposing such a simple constraint would be antithetical to the objective of accomplishing cross-domain image translation. Inspired by the latest findings in the domain of contrastive learning [60], we propose a novel contrastive constraint loss that focuses on local regions. The calculation flow for contrastive constraint loss is shown in Fig. 4. At first, the image of the source domain and its corresponding image of pseudotarget domain are input to the encoder E_G to obtain the layer-specific feature maps, denoted as F_S^i and F_P^i, respectively, where $i = 1, \ldots, 4$. Specifically, we select the feature maps of the first downsampling-convolution layer, the second downsampling-convolution layer, the residual block, and the last layer of E_G as shown in Fig. 2. In particular, as an option, we can extract more feature maps from the residual block. The prerequisite is that this residual block contains more layers. The feature map of each layer passes through a 1×1 convolution layer to reduce its dimension. Then, N vectors are selected at the corresponding positions in F_S^i and F_P^i, respectively, denoted as S_c^j and P_c^j, where $j = 1, \ldots, N$. Next, for each vector, its four neighbors are selected, denoted as $S_n^{j,k}$ and $P_n^{j,k}$, where $k = 1, \ldots, 4$. To measure the relationship between the central vector and its neighboring vectors, we use the following mathematical formula:

$$d_s^{j,k} = \text{sigmoid}\left(S_c^j - S_n^{j,k}\right), \quad d_p^{j,k} = \text{sigmoid}\left(P_c^j - P_n^{j,k}\right) \quad (2)$$

where the sigmoid function is specifically engineered to normalize the vector differences to fall within the range of 0 to 1. Its implementation enables the standardization of the vector differences, ensuring that they correspond to a normalized, scientifically meaningful scale. Based on the idea of contrastive learning, the potential space exhibits a higher correlation between $d_s^{j,k}$ and $d_p^{j,k}$ when compared with other random pairs. Ultimately the task can be abstracted as an $4 \times N$ classification problem. The objective function is a softmax-cross-entropy loss of the following form:

$$\mathcal{L}_c = \sum_i^M \sum_j^N \sum_k^4 -\log\left[\frac{\exp\left(d_s^{j,k} \cdot d_p^{j,k}/\tau\right)}{\sum_m^N \sum_n^4 \exp\left(d_s^{m,n} \cdot d_p^{m,n}/\tau\right)}\right] \quad (3)$$

where τ is the temperature coefficient, following the setting in MoCo [61], $\tau = 0.07$, M is the number of feature layers used to contrastive constraint.

D. Frequency Domain Constraint

Analyzing the contrastive constraint mentioned in the previous section, we find that it is controlling $\tilde{Y}$ and X to have similar content on a local region. In order to impose a direct constraint on the two in the global region, we propose an approach that involves approximating frequency information to preserve identity features. Specifically, we decompose each image into its respective high-frequency and low-frequency components via Gaussian kernels in the pixel

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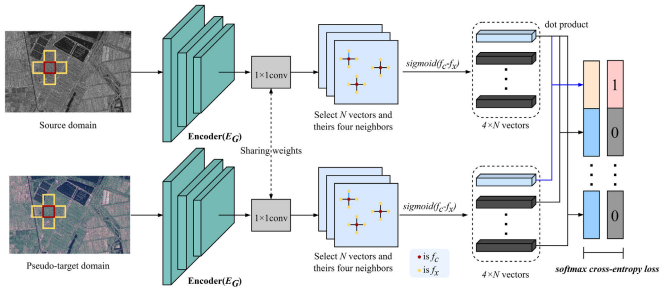


Fig. 4. Work flow of contrastive constraint. The red dot f_c represents the N vectors selected from F_S^i and F_P^i, which are denoted as S_c^j and P_c^j in the text. The yellow dot f_x represents the four neighbors of f_c, which are denoted as $S_n^{j,k}$ and $P_n^{j,k}$ in the text.

space. Subsequently, we minimize the disparity between the high-frequency components of the source domain image X and the pseudo-target domain image $\tilde{Y}$. The high-frequency components of the images are calculated as: $X_H = gray(X) - (gray(X))_L$, where the gray function converts a color image into a grayscale image to eliminate stylistic information deemed immaterial for preserving contents. The acquired high-frequency image X_H contains sharp edges. In addition, $\tilde{Y}_H$ is calculated in the identical way. The loss function in pixel space is as follows:

$$\mathcal{L}_{\text{pix}} = \mathbb{E}_{x\sim X,\tilde{y}\sim\tilde{Y}}\left[\left\|x_H - \tilde{y}_H\right\|_1\right]. \tag{4}$$

In Fourier spectral space, after the Fast Fourier Transform of the image, the consistency is adjusted directly in the frequency domain to ensure that X and $\tilde{Y}$ have similar high-frequency spectra. The specific formula for the Discrete Fourier Transform acting on an image I of size $H \times W$ is as follows:

$$\mathcal{F}(I)(u,v) = \frac{1}{HW}\sum_{h=0}^{H-1}\sum_{w=0}^{W-1} e^{-2\pi i\left(\frac{hu}{H}+\frac{wv}{W}\right)} \cdot I(h,w) \tag{5}$$

where $u = 0,\ldots,H-1$, $v = 0,\ldots,W-1$. The loss function of the spectral space is as follows:

$$\mathcal{L}_{\text{fft}} = \mathbb{E}_{x\sim X,\tilde{y}\sim\tilde{Y}}\left[\left\|\mathcal{F}_H(x) - \mathcal{F}_H\left(\tilde{y}\right)\right\|_1\right] \tag{6}$$

where $\mathcal{F}_H(x) = \mathcal{F}(x)\odot M_H$, M_H is the high frequency mask and it can retain high-frequency information. The specific formula for M_H is as follows:

$$M_H(u,v) = \begin{cases} 0, & (u-u_c)^2 + (v-v_c)^2 \le r^2 \\ 1, & \text{otherwise} \end{cases} \tag{7}$$

where u_c and v_c are the center coordinates of the spectrum, $r \in [0, \min((H/2),(W/2))]$.

By utilizing the proposed frequency information approximation technique, our model achieves greater effectiveness and efficiency in maintaining the identity features of the source domain images.

E. Total Loss Function

The loss function featured in this framework is expressed as a weighted sum, as follows:

$$\mathcal{L} = \mathcal{L}_{\text{GAN}} + \lambda_c\mathcal{L}_c + \lambda_p\left(\mathcal{L}_{\text{pix}} + \mathcal{L}_{\text{fft}}\right) \tag{8}$$

TABLE I
DETAILED INFORMATION OF FIVE DATASETS

Name	Date	Size	Type
Shuguang	June 2008 -Sept. 2012	593×921	SAR/RGB
Italy	Sept. 1995 -July 1996	412×300	Near Infrared/RGB
Texas	Sept. 2011 -Octo. 2011	1534×808	Multispectral/Multispectral
California	Janu. 2017 -Febr. 2017	3500×2000	Multispectral/SAR
France	May 2012 -July 2013	2000×2000	RGB/RGB

where weights λ_c and λ_p are used to balance the effect of the contrastive constraint and the frequency domain constraint on the translation results, respectively.

F. Change Detection Pipeline

In this article, a generator and a multiscale discriminator are trained without labels, and then the trained generator is used to obtain a pseudo-target domain image $\tilde{Y}$. The target domain and source domain images are exchanged, i.e., another pseudo-target domain image $\tilde{X}$ can be obtained by training again with X as the target domain. The two sets of approximately homogeneous images$(X, \tilde{X})$, $(Y, \tilde{Y})$ can be used as the input of the homogeneous change detection algorithm, and the results are obtained after filtering and threshold segmentation.

Finally, we calculate the difference map in each domain according to the following formula:

$$\text{diff} = \left(\left\|X - \tilde{X}\right\|_2 + \left\|Y - \tilde{Y}\right\|_2\right)/2. \tag{9}$$

By doing so we can use the information from both domains to get more accurate information about the changes. After that, we adopted the same filtering algorithm and segmentation algorithm as in the literature [33], [35], since our approach focuses on the translation part. This same operation renders our approach more suitable for comparison with existing methods.

IV. EXPERIMENTS

In this article, five heterogeneous remote sensing datasets are introduced in Section IV-A. And their details are reported in Table I. To cover the same event regions, all datasets were resampled, co-aligned, and cropped. Section IV-B provides the details of our experimental setup. In Section IV-C, we qualitatively compare the proposed method with several methods obtained from other HICD literature works on each dataset separately. Also, we provide the qualitative results of the translation on each dataset as well as the results of the change detection. In this section, we use overall accuracy (OA), F1 scores, and Cohen's Kappa Coefficient (KC) in the evaluation metrics for quantitative outcomes. For a more detailed

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Algorithm 1 MDCTNet Model

Input: Remote sensing images X and Y from different times, hyperparameters λ_c, λ_p, the temperature coefficient τ, the number of vectors extracted from each layer of the feature map in the contrastive constraint N, the Gaussian kernel size k_{size}, the high-frequency mask $Mask_H$

Output: The results of change detection

Input patches X into encoder E_G;
Obtain the feature map of each layer of the encoder F_S^i, where $i = 1 \cdots 4$;
Input the final layer of encoder output F_S^4 to the generator G;
Obtain generated pseudo-target domain data $\tilde{Y}$;
Input generated data $\tilde{Y}$ and target domain data Y into multiscale discriminator D_i, where $i = 1, 2, 3$;
Update the parameters of multiscale discriminator D to maximize $\mathcal{L}_{GAN}$;
Input generated data $\tilde{y}$ into encoder E_G;
Obtain the feature map of each layer of the encoder F_P^i, where $i = 1 \cdots 4$;
Extract N vectors from the reduced-dimensional F_S^i and F_P^i, respectively;
Obtain contrastive constraint loss $\mathcal{L}_c$ by Equation (2) and Equation (3);
Obtain frequency domain constraint loss $\mathcal{L}_{pix} + \mathcal{L}_{fft}$ by Equation (4) and Equation (6);
Update the parameters of encoder E_G and generator G to minimize Equation (8);
Swap the target and source domain and repeat the above process to get $\tilde{X}$;
Use X, $\tilde{X}$, Y, and $\tilde{Y}$ to obtain the results of change detection.



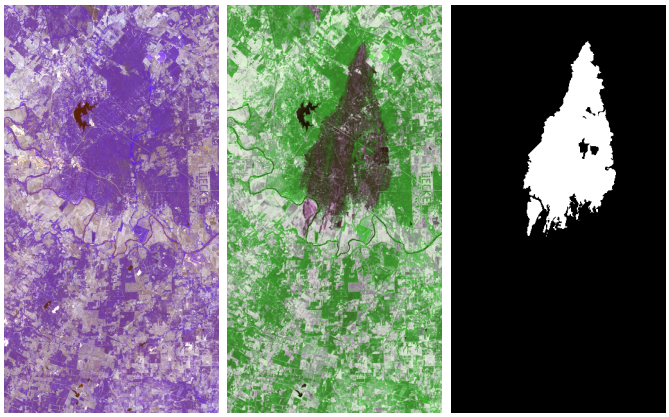
(a) (b) (c)

Fig. 5. Shuguang village dataset. (a) SAR (t_1). (b) RGB (t_2). (c) Ground truth.



(a) (b) (c)

Fig. 6. Italy dataset. (a) Near Infrared (t_1). (b) RGB (t_2). (c) Ground truth.



(a) (b) (c)

Fig. 7. Texas dataset. (a) Multispectral (t_1). (b) Multispectral (t_2). (c) Ground truth.

comparison, we added true positive (TP), true negative (TN), FP, and false negative (FN) to the already available evaluation metrics. Finally, in Section IV-D to highlight the contribution of each component in our methodology, an ablation study has been conducted on the Shuguang village dataset.

A. Datasets

1) Shuguang Village: The SAR image depicted in Fig. 5(a) and the coregistered optical image illustrated in Fig. 5(b) were acquired in June 2008 and September 2012, respectively, within the proximity of Shuguang Village, located in Dongying City, China. Both images exhibit a resolution of 8 m and consist of 593 × 921 pixels. The farmland variability is evident in the ground truth image displayed in Fig. 5(c). And the data source used in this article is from https://github.com/yulisun/INLPG/tree/master/datasets.

2) Italy: The source of this dataset is from this work [18]. The Landsat 5 images presented in Fig. 6, namely, (a) and (b), depict a 412 × 300 pixel area. The near-infrared (NIR) band image in (a) was acquired in September 1995, while the red, green, and blue (RGB) band image in (b) was obtained in July 1996 from the same region. The variation of the two images characterizes a lake flooding that occurred in Italy.

3) Texas: In September-October 2011, a forest fire occurred in Bastrop County, Texas. Landsat 5 TM captured an image of the area before the disaster [see Fig. 7(a)], while the Earth Observation-1 Advanced Land Imager (EO-1 ALI) acquired an image after the disaster [see Fig. 7(b)]. The two images presented in the study are captured through multispectral optical imaging with pixel dimensions of 1534 × 808 and comprise 7 and 10 channels, respectively. It is noteworthy that the two imaging platforms function at different spectral resolutions and utilize spectral bands that do not entirely overlap, making them heterogeneous. Volpi et al. [62] provided the ground truth of the region of variation displayed in Fig. 7(c).

4) California: On January 5, 2017, Landsat 8 captured images covering the area of Sacramento, Yuba, and Sutter counties in California. As shown in Fig. 8(a), in addition to the RGB channels, the Landsat 8 satellite's multispectral sensor furnishes eight additional channels spanning from deep blue to long-wave infrared. On February 18, 2017, the region was later affected by flooding, as shown in Fig. 8(b). The image was acquired by Sentinel-1a3 with the ratio between the two intensities of SAR polarization VV and VH as the third channel. This type of difference introduces a more

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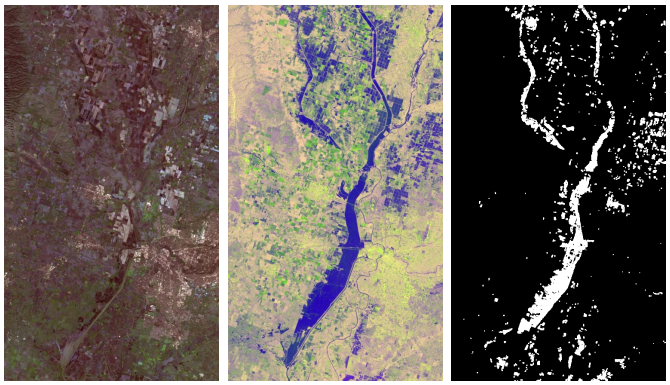


Fig. 8. California dataset. (a) Multispectral (t_1). (b) SAR (t_2). (c) Ground truth.

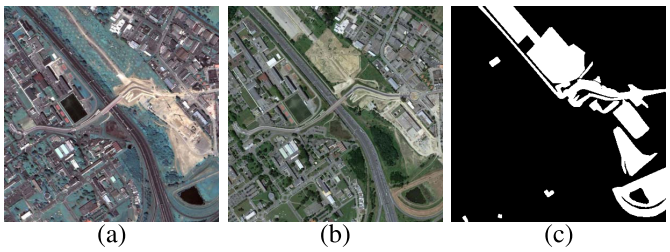


Fig. 9. France dataset. (a) RGB (t_1). (b) RGB (t_2). (c) Ground truth.

severe heterogeneity as we compare the multispectral and SAR images. The image pixel labels presented in Fig. 8(c) are attributed to Luppino et al. [26]. The original image size is 3500 × 2000 pixels. However, before the training process, downsampling was performed, resulting in an image size of 875 × 500 pixels.

5) France: The last dataset incorporates two distinct RGB images obtained by the Pleiades and WorldView 2 satellites, individually depicted in Fig. 9(a) and (b). It shows the events of the progress of the road construction work in Toulouse, France, between May 2012 and July 2013. The pixel labels in Fig. 9(c) depict this progress. We followed the work in [18] and reduced the pixels from 2000 × 2000 to 400 × 400. Although both images are RGB, they have slightly different spectral bands and are less heterogeneous compared to the previously mentioned datasets.

B. Implementation Setup

For our proposed framework, the translation phase consists of a fully convolutional network with a fully end-to-end training process. Among them, the structures of the encoder and discriminator have been described in detail in Section III-B, while for the generator the architecture is derived from the setup in NICE-GAN, using a six-layer residual network with adaptive layer instance normalization. Adam with a learning rate of 0.0001 is chosen as the optimizer. Since the heterogeneous image translation task is essentially a single image translation, one 96 × 96 pixel image patch is randomly selected at each iter and randomly enhanced (flipped or inverted). Doing so allows the network to be trained effectively with limited data. For the loss weights we set them

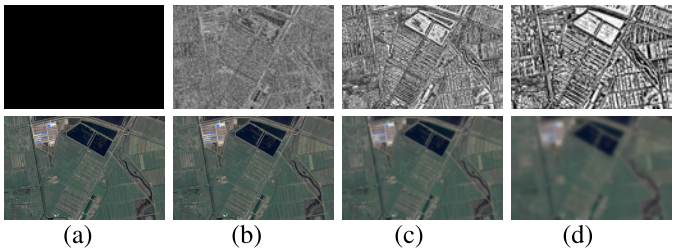


Fig. 10. Visualization of the effect of decomposing the original image into high-frequency (lower) and low-frequency (upper) components using Gaussian kernels of different sizes. (a) $k = 1$. (b) $k = 5$. (c) $k = 21$. (d) $k = 61$.

in the subsequent experiments: $\lambda_c = 10$, $\lambda_p = 1$. In particular, we use $\lambda_c = 20$ to get a better result on the Texas dataset. When choosing the size of the Gaussian kernel, we reflect the impact of the Gaussian kernel by the simple experiment in Fig. 10. Specifically, it is clear from the figure that a large kernel causes severe distortion in the separated low-frequency image, while a small kernel makes the high-frequency image miss a large amount of information. Therefore, the kernel size $k = 21$ was chosen for the vast majority of our experiments, while on the Texas dataset $k = 31$. This is due to the more complex ground cover of the Texas dataset.

In addition, all experimentation related to the proposed method was executed on a server outfitted with Intel Xeon Silver 4210R CPU @ 2.4 GHz and NVIDIA Geforce RTX 3080 GPU. Deep learning environment is Python 3.7 with PyTorch 1.12, cuda version 11.3, and cudnn version 8.3.

C. Comparison Experiments

The proposed method was compared with eight deep learning-based methods on these five datasets. Niu et al. [31] proposed a conditional adversarial network (CAN) for unsupervised heterogeneous CD tasks, whereas Liu et al. [30] introduced the SCCN for the same purpose. Both these models employ convolutional neural networks to tackle HICD tasks and are significant contributions to the field. Meanwhile, ACE-Net, X-Net, and CAAs are the latest approaches proposed by Luppino et al. [33], [35] and have shown good performance. However, due to the lack of experiments on certain datasets by the above methods, we supplemented the experiments with the code provided in this repository https://github.com/llu025/Heterogeneous_CD. This repository is mentioned in the literature [33]. Since there have been article [35] that has utilized this code to support their arguments, we think it is reasonable for us to do the same. The M-UNet [63] framework is a **supervised change detection** method and does not use a generative structure. This method directly classifies heterogeneous image patches directly during training. The DTCDN [34] uses a deep translation network to reduce the difference between optical images and SAR images and then imports the translation results into a **supervised change detection** network. In addition, MTCDN [57] is improved by DTCDN. MTCDN is a **supervised** multi-task learning framework that combines image translation and change detection tasks. Image translation is responsible for unifying two images into the same feature domain. Recently,

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UCDFormer [58] employs a lightweight transformer-driven approach for image translation, aiming to alleviate domain shift efficiently in real-time.

Comparing the proposed MDCTNet with the above methods, it can be found that MDCTNet uses adversarial training similar to CAN and ACE-Net. Although GANs are unstable, we find that the dominant approach for image-to-image conversion tasks is still mostly in the GAN paradigm and has good results. Both SCCN and CCA use a code-alignment scheme, which has proven to be effective, but MDCTNet is not applicable as each training is a one-way conversion. We can include it in subsequent improvement schemes. X-Net, ACE-Net, and CAA all use a similar cycle consistency and weighted image translation strategy, but these methods rely on the acquisition of a priori information to weight the target function. In our proposed method a completely different strategy is used instead of cycle consistency and weighted image translation. It is important to note that, for supervised methods DTCDN and MTCDN, we do not employ the deep supervised classification network as described in the original articles. Instead, the images generated by the image translation network of DTCDN and MTCDN are directly inputted into the unsupervised change detection pipeline mentioned in our method. This decision was made to ensure the fairness of the experimental results. As for the other supervised method, M-UNet, since M-UNet does not have a translation-based structure, we only simply give its experimental results without a rigorous comparison. In the experiments below, we provide the results obtained on the five datasets by MDCTNet and the above method. It can be seen that the contrastive constraint and the frequency domain constraint lead to better results.

1) Results on Shuguang Village Dataset: Table II shows the quantitative results of the proposed MDCTNet on the Shuguang village dataset with the five HICD algorithms. Analysis of the data in the table shows that our method achieves a higher OA than other methods. While the performance of MDCTNet is similar to the OA of X-Net, it boasts superior F1 and KC metric levels. For example, our method surpasses its most successful counterpart, not only in terms of F1 score by 16.43%, but also in KC by 16.91%. Since the pixel points that change in the change detection dataset are in the minority, F1 score, and KC are more reasonable metrics to evaluate than OA. Therefore, this further suggests that our method is more effective than the other methods on the Shuguang village dataset.

2) Results on Italy Dataset: The results of the evaluation metrics garnered from the Italy dataset are shown in Table III. Again, our method outperforms the state-of-the-art method in OA, F1, and KC by 0.94%, 1.21%, and 1.61%, respectively. It is worth noting that SCCN, CAN, X-Net, and ACE-Net do not give the values of their KC and F1 scores on this dataset in the article, so we used the code provided by [33] to obtain the full results.

It is noted that SCCN and ACE-Net achieve poorer results on this dataset. Based on the previous analysis of these methods, it is concluded that SCCN relies only on code-space alignment, which lacks more constraints. ACE-Net uses the well-known cycle consistency, but it uses adversarial training methods, which brings some problems.

TABLE II
PERFORMANCE EVALUATION OF VARIOUS METHODS ON THE SHUGUANG VILLAGE DATASET

Method	TP	TN	FP	FN	OA(%)	F1(%)	KC(%)
SCCN [30]	20644	472914	46627	4455	90.62	44.70	40.72
CAN [31]	10972	502482	17059	14127	94.27	41.30	38.30
X-Net [33]	20258	502924	16617	4841	96.06	65.38	63.37
ACE-Net [33]	20292	494964	24577	4807	94.60	58.00	55.37
CAA [35]	20747	493356	26185	4352	94.39	57.61	54.90
UCDFormer [58]	13029	519282	1772	12070	97.47	65.31	64.08
MDCTNet	19469	518029	3025	5630	**98.41**	**81.81**	**80.99**
DTCDN [34]	15986	505541	15513	9113	95.49	56.49	54.14
M-UNet [63]	24070	490309	30745	1029	94.18	60.24	57.56
MTCDN [57]	16502	500541	20513	8597	94.67	53.13	50.42

TABLE III
PERFORMANCE EVALUATION OF VARIOUS METHODS ON THE ITALY DATASET

Method	TP	TN	FP	FN	OA(%)	F1(%)	KC(%)
SCCN [30]	6164	6164	10959	1462	89.95	49.81	45.13
CAN [31]	4228	111660	4314	3398	93.76	52.30	48.97
X-Net [33]	3651	108701	7273	3975	90.90	39.36	34.61
ACE-Net [33]	4097	108058	7616	3529	90.74	41.72	36.97
CAA [35]	5833	109915	6059	1793	93.65	59.77	56.50
UCDFormer [58]	6142	110294	5680	1484	94.20	63.16	60.18
MDCTNet	5427	112166	3808	2199	**95.14**	**64.37**	**61.79**
DTCDN [34]	2577	102663	13311	5049	85.15	21.92	14.82
M-UNet [63]	6808	110833	5141	818	95.18	69.56	67.08
MTCDN [57]	3017	110955	5019	4609	92.21	38.53	34.37

3) Results on Texas Dataset: Table IV shows the quantitative results of several methods on the Texas dataset. Not surprisingly, our method again outperforms the others in all metrics. Specifically, MDCTNet improves by 0.48%, 2.49%, and 2.76% on OA, F1, and KC, respectively. The dataset under consideration comprises a significant number of channels, which translates to an increased capability of discerning complex ground changes compared to datasets with a smaller number of channels. SCCN and CAN, however, are significantly worse than the other methods. One is mainly attributed to fewer constraints, while the other is more related to their relatively simpler network structure which is insufficient to accommodate larger quantities of information in this dataset.

4) Results on California Dataset: The results for each indicator obtained on the California dataset are summarized in Table V. In contrast to the previous results, although our method is slightly behind the state-of-the-art in OA, it improves by 0.95% and 0.92% in F1 and KC, respectively. This does not negate the validity of our method, because our

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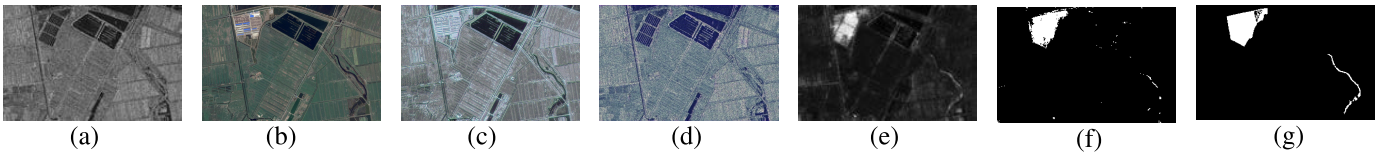


Fig. 11. Results of the translation on the Shuguang village dataset and the final change detection results. (a) X. (b) Y. (c) $\tilde{X}$. (d) $\tilde{Y}$. (e) Filtered. (f) Change map. (g) Ground truth.

TABLE IV
PERFORMANCE EVALUATION OF VARIOUS METHODS ON THE TEXAS DATASET

Method	TP	TN	FP	FN	OA(%)	F1(%)	KC(%)
SCCN [30]	112559	1014151	93452	19310	90.90	66.63	61.65
CAN [31]	63615	1084411	23192	68254	92.62	58.18	54.32
X-Net [33]	108352	1083460	17356	30304	96.15	81.97	79.83
ACE-Net [33]	122832	1079832	20984	15824	97.03	86.97	85.29
CAA [35]	115788	1094259	13344	16081	97.63	88.73	87.40
UCDFormer [58]	38307	1083974	23629	93562	90.55	39.53	35.12
MDCTNet	121218	1094907	12696	10651	**98.11**	**91.22**	**90.16**
DTCDN [34]	69564	1088561	19042	62305	93.44	63.10	59.63
M-UNet [63]	122644	1103343	4260	9225	98.91	94.76	94.18
MTCDN [57]	83249	1086628	20975	48620	94.39	70.52	67.46

TABLE V
PERFORMANCE EVALUATION OF VARIOUS METHODS ON THE CALIFORNIA DATASET

Method	TP	TN	FP	FN	OA(%)	F1(%)	KC(%)
SCCN [30]	25350	369250	31284	11116	90.30	54.45	49.31
CAN [31]	12730	386241	14293	23736	91.30	40.10	35.52
X-Net [33]	18979	379310	21224	17487	91.14	49.51	44.67
ACE-Net [33]	18027	377570	22964	18439	90.52	46.55	41.37
CAA [35]	18939	387379	13155	17527	**92.98**	55.25	51.45
UCDFormer [58]	19617	385890	15077	16916	92.69	55.08	51.11
MDCTNet	19746	386963	14004	16787	92.96	**56.20**	**52.37**
DTCDN [34]	6773	371172	29795	29760	86.39	18.53	11.10
M-UNet [63]	25940	372234	28733	10593	91.01	56.88	52.09
MTCDN [57]	15769	364343	20764	1064	86.88	35.47	28.42

TABLE VI
PERFORMANCE EVALUATION OF VARIOUS METHODS ON THE FRANCE DATASET

Method	TP	TN	FP	FN	OA(%)	F1(%)	KC(%)
SCCN [30]	7480	121997	10251	20272	80.92	32.89	22.40
CAN [31]	4530	127014	5234	23222	82.21	24.15	16.62
X-Net [33]	9199	127348	4900	18553	85.34	43.96	36.55
ACE-Net [33]	8236	125752	6496	19516	83.74	38.77	30.40
CAA [35]	9490	127743	4505	18262	85.77	45.46	38.29
UCDFormer [58]	10838	115107	17141	16914	78.72	38.89	6.01
MDCTNet	10133	128254	3994	17619	**86.49**	**48.39**	**41.55**
DTCDN [34]	7344	125320	6928	20408	82.92	34.95	26.26
M-UNet [63]	24990	118928	13320	2762	89.95	75.66	69.53
MTCDN [57]	3321	128908	3340	24431	82.64	19.30	13.49

method still has a large improvement in all metrics compared to the methods except CAA.

It is worth mentioning that in the literature [35] it is mentioned that because this dataset involves a more severe heterogeneity: SAR images versus multispectral images - the difference in channels makes it not easy to achieve the regression from less to more variables. To address this issue, we perform PCA [64] downscaling on the multispectral images when making the training set and use the downscaled images instead of the original images for training as well as for the final change detection.

5) Results on France Dataset: Table VI provides the results of several methods on the France dataset. Once again, our proposed MDCTNet achieves optimal outcomes. It should be noted that our comparison is limited to some methods based on deep learning. Some traditional methods such as [18] and [27] have also achieved good results.

The figures presented, namely Figs. 11 through 15, display the visualization outcomes of the translation procedure employing MDCTNet across all the datasets analyzed in this section. X and Y represent the original image, $\tilde{X}$ and $\tilde{Y}$ represent the converted image, Filtered represents the gray map obtained after filtering algorithm for the diff calculated by (9). Change map and Ground truth represent the results predicted by the model and the real change detection results, respectively, and they are binary maps, where the white pixels represent the regions that have changed and the black pixels represent the regions that have not changed. The collective results showcase that the data originating from the source domain are transformed in a meaningful manner, simulating the pseudo-target domain. The similarity of visual features between the pseudo-target domain image and the target domain image image is evident. Also, it can be observed that on the California dataset, the "LUECKE" character is perfectly preserved in the translation, although the ground cover is complex. This also illustrates the excellent performance of our method in retaining content features. Finally, for a more visual comparison of MDCTNet with other methods, Figs. 16–20 provide qualitative results for each method on five datasets (**TP: white; TN: black; FP: green; FN: red**). The quantitative and qualitative results jointly demonstrate that the proposed method carries a lower false alarm rate, in turn effectively tackling the heterogeneity-triggered false

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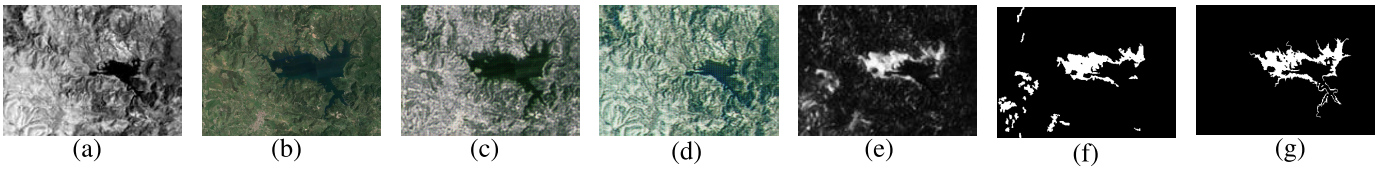


Fig. 12. Results of the translation on Italy dataset and the final change detection results. (a) X. (b) Y. (c) $\tilde{X}$. (d) $\tilde{Y}$. (e) Filtered. (f) Change map. (g) Ground truth.

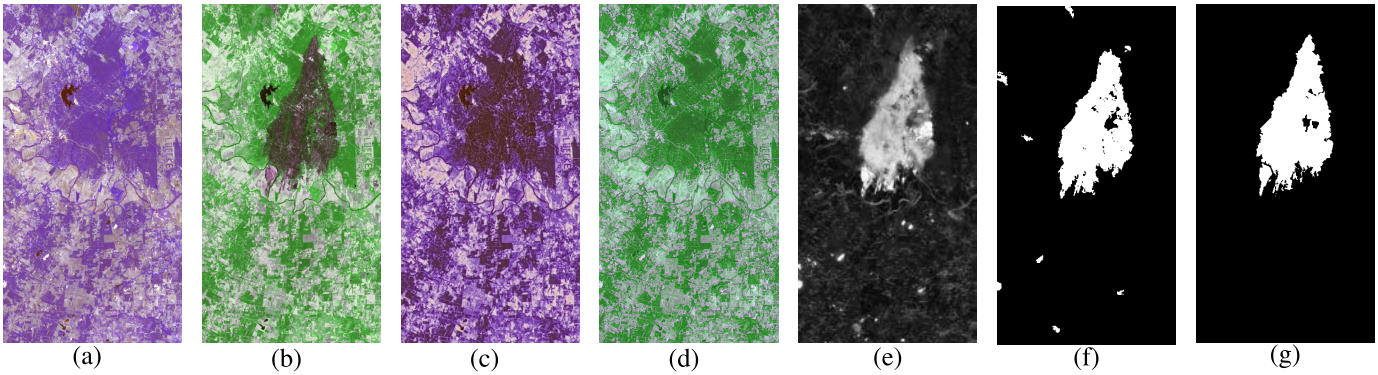


Fig. 13. Results of the translation on Texas dataset and the final change detection results. (a) X. (b) Y. (c) $\tilde{X}$. (d) $\tilde{Y}$. (e) Filtered. (f) Change map. (g) Ground truth.

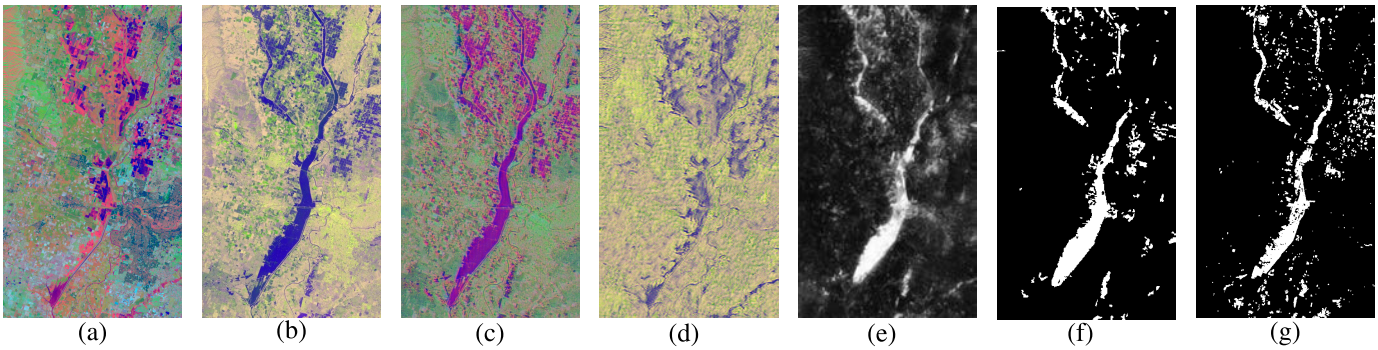


Fig. 14. Results of the translation on California dataset and the final change detection results. Here, X represents the result of PCA downscaling of the original multispectral image. (a) X. (b) Y. (c) $\tilde{X}$. (d) $\tilde{Y}$. (e) Filtered. (f) Change map. (g) Ground truth.



Fig. 15. Results of the translation on France dataset and the final change detection results. (a) X. (b) Y. (c) $\tilde{X}$. (d) $\tilde{Y}$. (e) Filtered. (f) Change map. (g) Ground truth.

alarm issue. Moreover, it is found that DTCDN has a high FN rate across datasets, which is due to the fact that the translation network in DTCDN does not explicitly constrain the content of the images. This further highlights the role of the two constraint modules in the proposed method.

D. Ablation Experiment

In order to analyze the contribution of each component of MDCTNet to the final results, we perform an ablation study on the Shuguang village dataset.

In Table VII, we provide MDCTNet and the results obtained for different configurations. Specifically, the MDCTNet has the following variants.

1) Baseline.
2) Only multiscale discriminator.
3) Only contrastive constraint.
4) Only frequency domain constraint.
5) MDCTNet without frequency domain constraint.
6) MDCTNet without contrastive constraint.
7) MDCTNet without multiscale discriminator.
8) Full MDCTNet.

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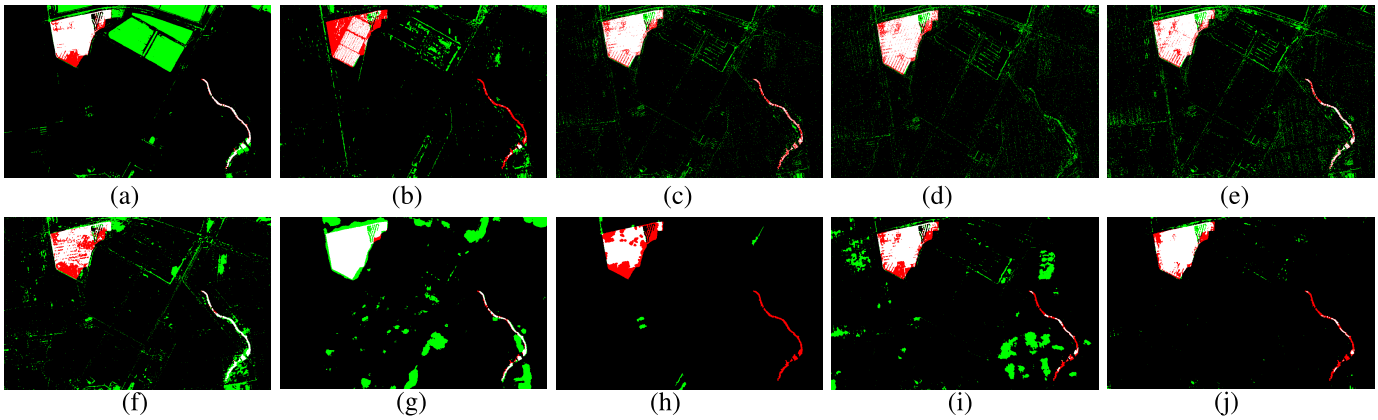


Fig. 16. Qualitative results for each method on the Shuguang village dataset. (a) SCCN. (b) CAN. (c) X-Net. (d) ACE-Net. (e) CAA. (f) DTCDN. (g) M-UNet. (h) UCDFormer. (i) MTCDN. (j) MDCTNet.

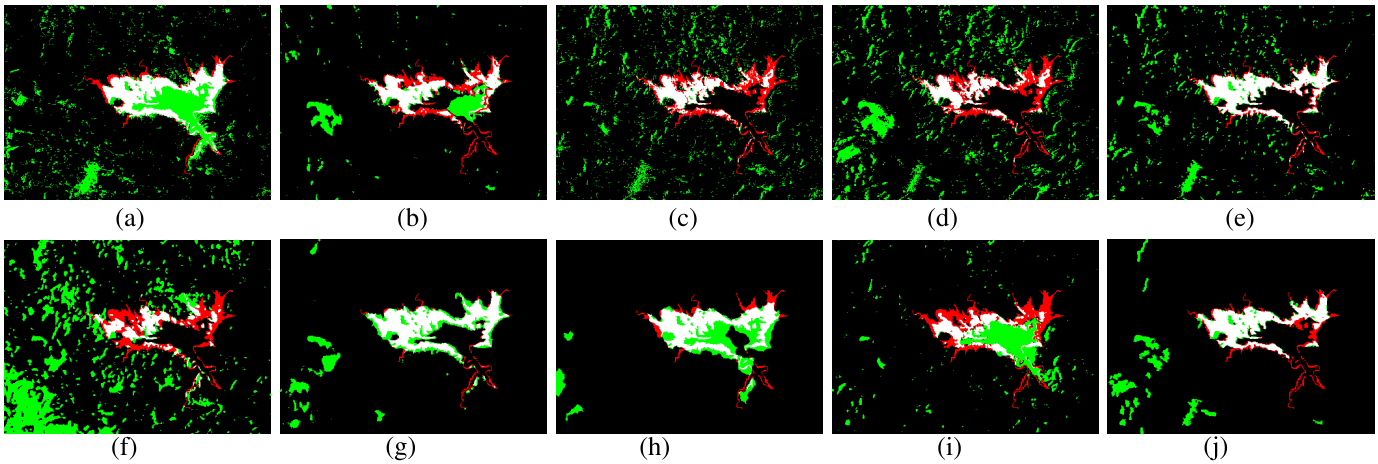


Fig. 17. Qualitative results for each method on the Italy dataset. (a) SCCN. (b) CAN. (c) X-Net. (d) ACE-Net. (e) CAA. (f) DTCDN. (g) M-UNet. (h) UCDFormer. (i) MTCDN. (j) MDCTNet.

TABLE VII

ABLATION STUDY OF THE SHUGUANG VILLAGE DATASET: RESULTS OBTAINED FOR VARIOUS CONFIGURATIONS. THE BEST RESULTS ARE SHOWN IN BOLD

Multi-scale discriminator	Contrastive constraint	Frequency domain constraint	TP	TN	FP	FN	OA(%)	F1(%)	KC(%)
✗	✗	✗	4227	478472	42582	20872	88.38	11.75	6.14
✓	✗	✗	20631	462579	58475	4468	88.47	39.59	35.06
✗	✓	✗	18339	518879	2175	6760	98.36	80.41	79.56
✗	✗	✓	8862	473337	47717	16237	88.29	21.70	16.37
✓	✓	✗	19193	517978	3076	5906	98.35	81.04	80.18
✓	✗	✓	**21448**	498290	22764	**3651**	95.16	61.89	69.51
✗	✓	✓	17753	**519287**	**1767**	7346	98.33	79.57	78.72
✓	✓	✓	19469	518029	3025	5630	**98.41**	**81.81**	**80.99**

The ablation experiment shows that each part of the proposed MDCTNet contributes to the final result. Comparing the latter three sets of experiments, it can be concluded that all three components contribute to the final results, specifically, the contrastive constraint is crucial. Analyzing the results of the first four variants in the table, it can be observed that the performance of the model becomes poor if only the multiscale discriminator or the frequency domain constraint is used. However, their overall results are still better than the unsupervised baseline. The variant using only frequency-domain constraints has a lower OA than the baseline model, but its F1 score and KC are still much higher than the baseline. Furthermore, the results of this study show that the accuracy achieved by the model relying solely on the contrastive constraint is comparable to that of the full MDCTNet. This finding underscores the value of the said

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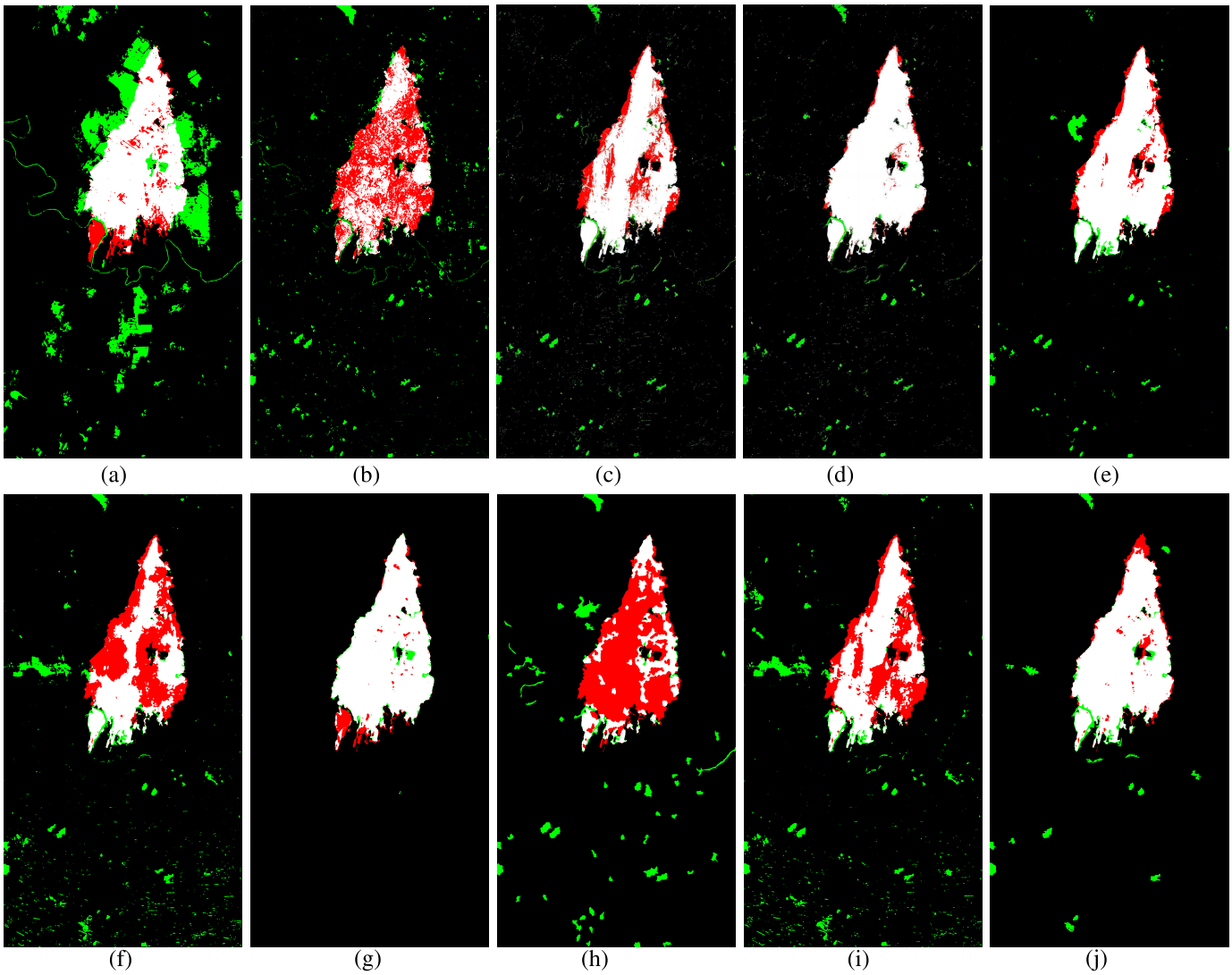


Fig. 18. Qualitative results for each method on the Texas dataset. (a) SCCN. (b) CAN. (c) X-Net. (d) ACE-Net. (e) CAA. (f) DTCDN. (g) M-UNet. (h) UCDFormer. (i) MTCDN. (j) MDCTNet.

TABLE VIII
HYPERPARAMETER SELECTION EXPERIMENT ON SHUGUANG VILLAGE DATASET

λ_c	λ_p	OA(%)	F1(%)	KC(%)
10	10	98.29	80.94	80.05
10	5	97.71	73.35	72.16
10	0.5	95.96	56.67	54.56
20	1	98.19	80.88	79.94
1	1	94.92	56.35	53.78
5	1	95.63	58.91	56.65
10	1	**98.41**	**81.81**	**80.99**

contrastive constraint. It is also interesting to note that on the incomplete variants (i.e., 0–6), the inclusion of multiscale discriminators or frequency domain limitations may not invariably result in an enhancement in the overall performance, but the addition of contrastive constraint does lead to a substantial performance improvement. For example, Upon comparing the third and fifth variants, it can be found that the inclusion of a multiscale discriminator results in an OA reduction of 0.01%, but comparing the fourth and sixth variants, it can be found a 6.87% increase in OA. Also, comparing the results obtained from variants 2 and 5, variants 4 and 7, the addition of the contrastive constraint improves the OA by 9.88% and 10.04%, respectively. Finally, by comparing the third, fifth, and seventh variants, it is found that adding either the multiscale discriminator or the frequency domain constraint alone brings about a slight performance degradation in the model with existing contrastive constraint. It should be believed these two components have a mutually complementing relationship, and require simultaneous presence to have a positive effect on the system. By comparing the second, fourth, and sixth variants, it is also found that the sixth variant with multiscale discriminator and frequency domain constraint gives significantly better results than variants (2 and 4) with only one of them.

In addition to verifying the effect of the above three components on the model, we also conducted experiments

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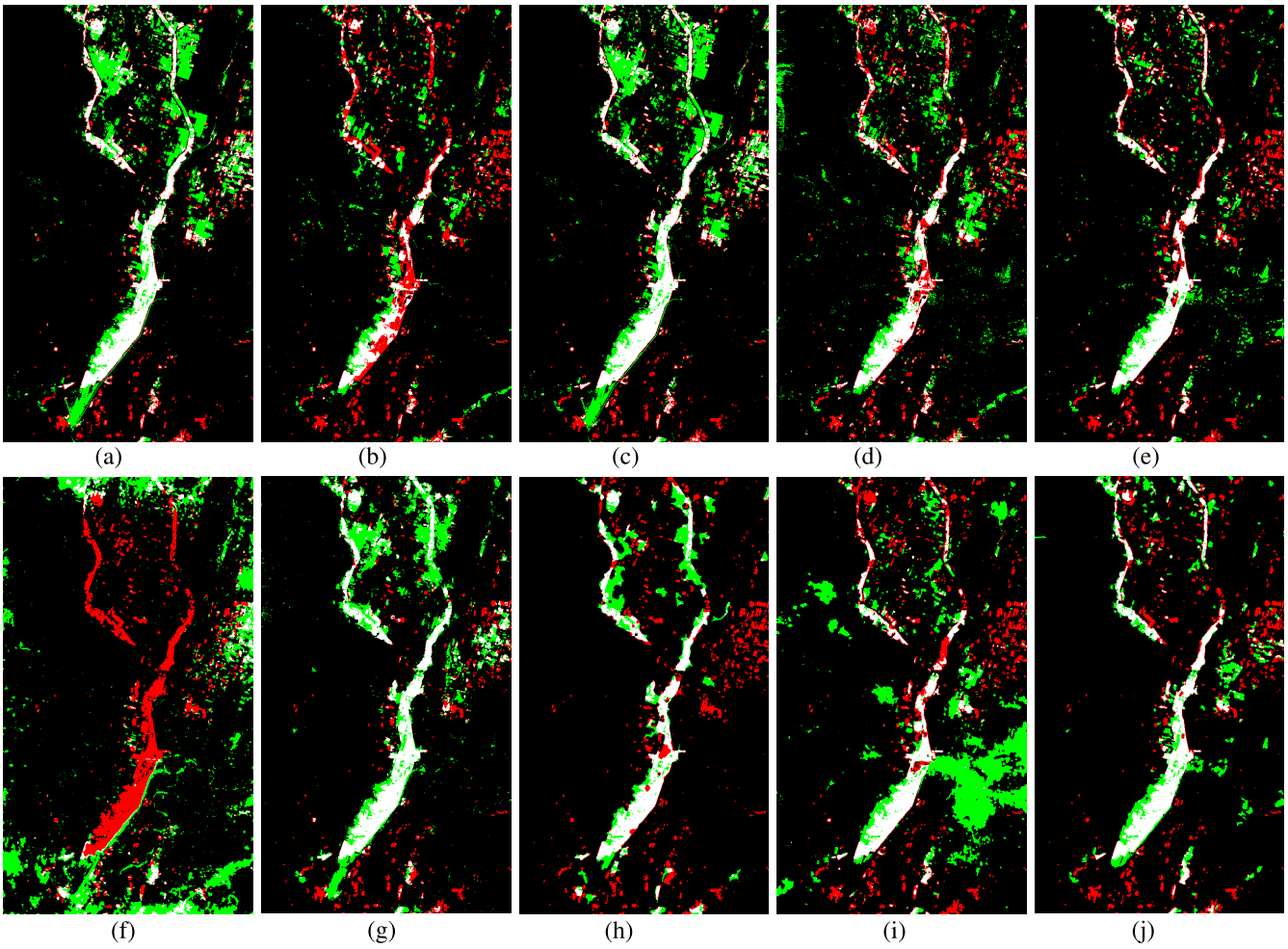


Fig. 19. Qualitative results for each method on the California dataset. (a) SCCN. (b) CAN. (c) X-Net. (d) ACE-Net. (e) CAA. (f) DTCDN. (g) M-UNet. (h) UCDFormer. (i) MTCDN. (j) MDCTNet.

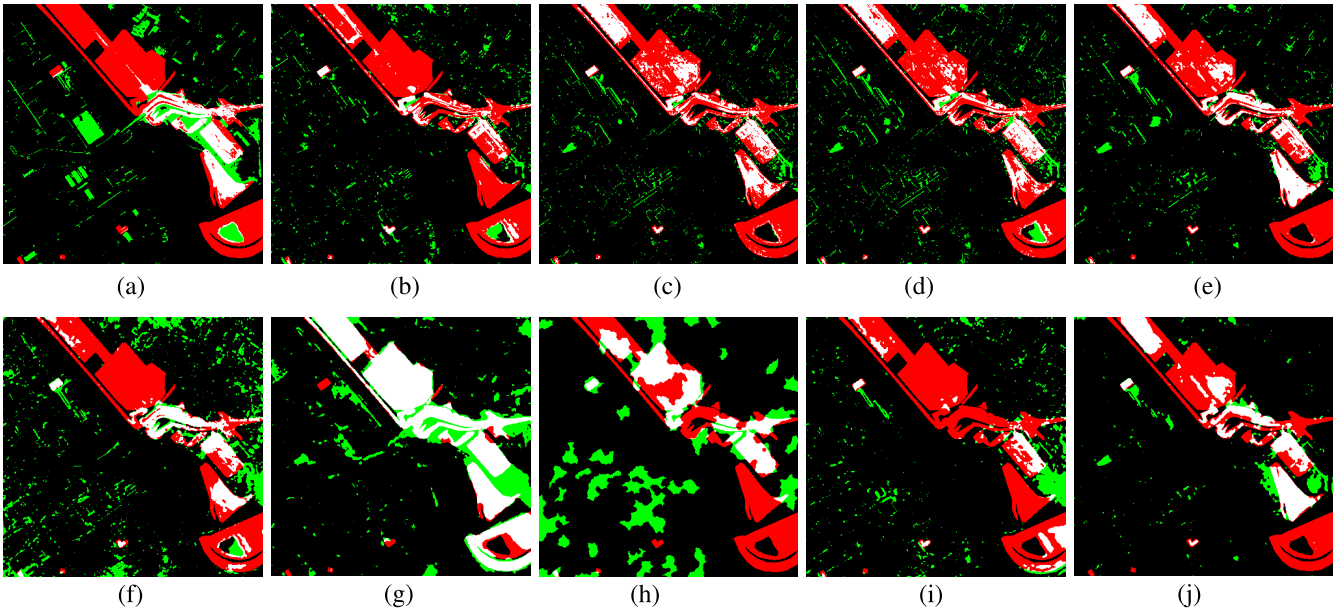


Fig. 20. Qualitative results for each method on the France dataset. (a) SCCN. (b) CAN. (c) X-Net. (d) ACE-Net. (e) CAA. (f) DTCDN. (g) M-UNet. (h) UCDFormer. (i) MTCDN. (j) MDCTNet.

on the effect of the values of the hyperparameters λ_c and λ_p on the results, which is shown in Table VIII. Indeed, we measured the initial values of the loss terms and then chose appropriate hyperparameters to balance them. The results of multiple experiments in Table VIII demonstrate that this is feasible.

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V. Conclusion

In this article, we propose an end-to-end MDCTNet for unsupervised HICD. The method uses an improved GAN to realistically generate the target domain image. To avoid the loss of change information during the translation process, it leverages contrastive learning to ensure the consistency of adjacent pixel relationships. Meanwhile, it employs high-frequency information consistency which preserves pivotal characteristics. The experimental results obtained from five heterogeneous datasets demonstrate that the proposed MDCTNet has achieved commendable outcomes. It is proved that the combination of contrastive constraint and frequency domain constraint further improves the accuracy of change detection.

The proposed MDCTNet employs PCA to reduce the dimensionality of the input images to a uniform dimension, resulting in a partial loss of information. As a future endeavor, we aim to investigate the possibility of applying other types of generative models such as normalizing flow models, variational autoencoder (VAE), and denoising diffusion probabilistic models.

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